

Skill Discovery Objective for flow models

The objective aims to maximise the mutual information between the state transitions (s, s') produced by a policy π , and a latent skill variable z , drawn from a distribution of skills $z \sim p(z)$.

The intuition for this choice of objective can be made more apparent if we consider the definition of mutual information,

$$I(s, s'; z | \Pi) = H(s, s' | \Pi) - (s, s' | z, \Pi)$$

Maximising the mutual information can be interpreted as maximising the marginal state-transition entropy $H(s, s' | \pi)$ induced by π , while minimising the conditional state-transition entropy produced by a particular skill z . In other words, the policy should learn a set of skills that produces diverse behaviours, while each individual skill should produce a distinct behaviour.

$$H(s, s' | \pi) \text{ induced by } \Pi$$

The above equation is intractable,

Approximate it by Variational Lower Bound:

By Symmetry:

$$I(s, s'; z | \Pi) = I(z; s, s' | \Pi)$$

$$I(s, s'; z | \Pi) = H(z) - H(z | s, s', \Pi)$$

$P(z)$ can still be truly defined in the flow model

$$Z_{i-1} \sim P_{i-1}(Z_{i-1})$$

$$Z_i = f_i(Z_{i-1})$$

$$Z_{i-1} = f_i^{-1}(Z_i)$$

$$P_i(Z_i) = P_{i-1}(f_i^{-1}(Z_i)) \left| \det \left(\frac{df_i}{dz_i} \right)^{-1} \right|$$

Let us convert the equation to be a function of Z so we can do inference with base distribution

$$P_i(Z_i) = P_{i-1}(f_i^{-1}(z_i)) \left| \det \left(\frac{df_i}{dz_i} \right)^{-1} \right|$$

By Inverse Function theorem,

$$P_i(Z_i) = P_{i-1}(z_i - 1) \left| \det \left(\frac{df_i}{dz_{i-1}} \right)^{-1} \right|$$

Jacobian property of invertible function,

$$P_i(Z_i) = P_{i-1}(z_i - 1) \left| \det \left(\frac{df_i}{dz_{i-1}} \right) \right|^{-1}$$

taking log on both side,

$$\log P_i(Z_i) = \log P_{i-1}(z_{i-1}) - \log \left| \det \left(\frac{df_i}{dz_{i-1}} \right) \right|$$

If $y = f(x)$ and $x = f^{-1}(y)$

$$\frac{df^{-1}(x)}{dy} = \frac{dx}{dy} = \left(\frac{dy}{dx} \right)^{-1} = \left(\frac{df(x)}{dx} \right)^{-1}$$

$$\det(M^{-1}) = (\det(M))^{-1}$$

as,

$$\det(M)(\det M^{-1}) = \det(M \cdot M^{-1}) = \det(I) = 1$$

Given such a chain of PDF's, we know the relation between each pair of consecutive variables.

We expand the equation of output x until we reach original distribution,

$$x = Z_k = f_k \cdot f_{k-1} \dots f_1(z_0)$$

$$\log p(x) = \log_{\Pi_k}(Z_k) = \log_{\Pi_{k-1}}(z_{k-1}) - \log \left| \det \frac{df_k}{dz_{k-1}} \right|$$

$$\log p(x) = \log_{\Pi_k}(Z_k) = \log_{\Pi_0}(z_0) - \sum_{i=1}^k \log \left| \det \frac{df_i}{dz_{i-1}} \right|$$

Now, to tractably approximate $H(z|s, s', \pi)$

We introduce variational approximation $q(z|s, s')$ which would be the inverse of the flow to reconstruct motion s_t ,

thus the surrogate objective becomes,

$$P_s(s) = \log(P_z(f(s))) + \log \left| \det \left(\frac{\partial f(s)}{\partial s^T} \right) \right|$$

$$\log(P_x x) = \log(P_z(f(s))) + \log \left| \det \left(\frac{\partial f(s)}{\partial s^T} \right) \right|$$

Reward would be negative less likelihood of chosen, that is,

$$r_t = -\log P_z(f(s)) - \log \left| \det \frac{\partial f(s)}{\partial s^t} \right|$$

The sequence of latents is constructed such that a latent z is repeated for multiple timesteps, before a new latent is sampled from $p(z)$ and repeated for multiple subsequent timesteps. This encourages the model to learn to transition between different skills.

Surrogate objective with diversity objection, and this surrogate objective along with diversity objective becomes,

$$\arg \max_{\pi} E_{P(z)} E_{P(\tau|\pi, z)} \left[\sum_{t=0}^{T-1} r^{\tau} (\log P_z(s)) \right] - w_{div} E_{d^{\pi}(s)} E_{z_1, z_2 \sim P(z)} \left[D_{KL} \frac{\pi(|s, z_1), \pi(\cdot | s)}{D_z(z_1, z_2)} \right]$$

The last term is the diversity objective, which stipulates that if two latents z_1 and z_2 are similar under a distance function D_z , then the policy should produce similar action distributions, as measured under the KL-divergence.

The Cosine distance between z_1 and z_2 ,

$$D_2(z_1, z_2) = \frac{1}{2}(1 - z_1^T - z_2)$$

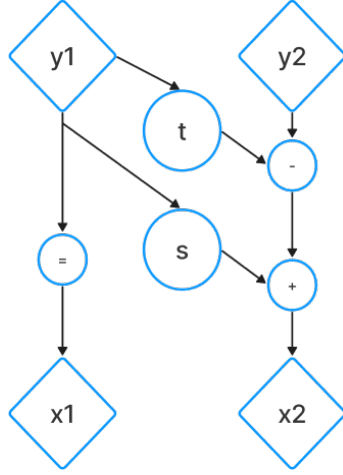


Fig 3.2: Backward propagation

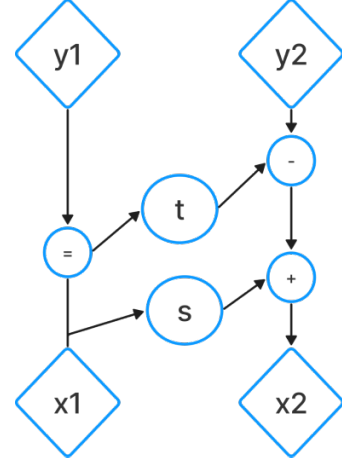


Fig 3.3: Forward propagation

and thus the reward at time step t becomes,

$$r_t = -\log(P_x(x))$$

$$r_t = -\log \pi_0(z_0) + \sum_{i=1}^k \log \left| \frac{dz_i}{dz_{i-1}} \right|$$